WhenWhereWhat: Every Place Has Its Moment, and Every Traveler Deserves to Find It

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***Abstract*—Most travel apps suggest places to visit in New York City, but they do not say when to visit them. Timing matters, because the same place can be crowded at one hour and quiet at another, and a view looks very different in the morning and at sunset. This paper describes the first phase of a planner that makes a day plan and suggests a time for each stop. Three public datasets are used: Foursquare check-ins from 2012 and 2013, Google business data for New York, and hourly pedestrian counts from the Brooklyn Bridge. The data is cleaned and joined in fifteen steps, two more steps prepare it for a preference form, and thirteen exploratory analyses are carried out. The results show that different types of places are busiest at different times. About two thirds of the check-ins are in Manhattan, nightlife and entertainment are more common on weekends, rain reduces bridge crowds by about three quarters, and the bridge is much quieter in the evening than in the afternoon. A strong sunset effect was expected, but a more careful test shows that it is very small. Finally, a basic test planner is built. It creates a main plan and a backup plan from the user's answers, and the problems in its results are noted as future work.**

***Index Terms*—itinerary planning, data cleaning, exploratory data analysis, check-in data, crowd patterns**

I. Introduction

Travel websites usually give lists such as "top 20 places to visit in New York". These lists are helpful, but they do not answer simple practical questions, for example "What time is the Brooklyn Bridge least crowded?" or "Which parks are pleasant in the evening?". In most cases a visitor learns these things only after reaching the place, often after standing in a long line or finding that the place is already closed.

The goal of this project is to help a traveler decide where to go and when to visit each place. The app is planned to work like this. The user logs in and fills out a short form about interests, preferred time of day, feelings about crowds, the pace of the day, budget and travel date. The app then returns a day plan in which every stop has a suggested time. If the user dislikes a stop, a backup plan is shown. The project is called WhenWhereWhat because the planner tries to answer three questions for every stop: where to go, when to go and what to do next.

A. Problem Statement

Given a traveler's answers to a short form and a travel date, the task is to build a one-day plan for New York City in which each stop matches the traveler's interests, is visited at a time when it is pleasant and open, and is followed by a sensible next stop. For this to be a data science problem and not only a user interface, every question in the form must be linked to something that can be measured in real data. For example, the answer "I like mornings" should be linked to the share of visits a place receives in the morning.

B. Objectives of Phase 1

This report describes Phase 1, where the data is prepared. The main objectives of this phase are:

- to clean three datasets that have very different formats and come from different years, and to join them together;
- to carry out thirteen analyses that study how visits change by hour, day, weather and sunset, and to use the results to guide the design;
- to build tables that link every form question to a feature in the data, and a simple test planner that checks whether all parts work together.

One result was different from what we first expected. The idea of visiting viewpoints just before sunset was expected to be the strongest part of the project, but the data supports it only weakly. This is discussed in Section VI-D.

II. Related Work

Building trip plans from social media data has been studied before. De Choudhury et al. used the time stamps of Flickr photos to estimate how long people stay at a place and where they go next, and used this information to build travel itineraries [1]. Lim et al. added each user's interests and typical visit durations to recommend personalized tours [2]. These works mainly focus on the order of places and on the time spent at each place.

The Foursquare dataset used here was released by Yang et al., who used it to study how user activities depend on place

TABLE I
DATASETS USED IN PHASE 1

Source	Raw rows	Years	Used for
Foursquare	227,428	2012–13	timing, sequences
Google Local	272,189	about 2021	names, ratings, hours
Brooklyn Bridge	16,057	2017–19	crowds, weather

and time [3]. The Google business data was collected by the McAuley lab at UC San Diego [4].

The works above do not use measured crowd counts or the sunset time of the travel date to choose when a place should be visited. This project attempts to add these two elements, and it also uses a short form instead of a long user history, so that a first-time visitor can still get a useful plan.

III. DATASETS

Three datasets are used in this project. A short summary is given in Table I.

A. Foursquare NYC Check-ins

A check-in is a record of a user telling the Foursquare app that they are at a certain place. The dataset [3] contains 227,428 check-ins by 1,083 users at 38,333 places, from April 2012 to February 2013. Each row has a user ID, a place ID, one of 251 place categories (such as "Park" or "Bar"), the location and the time. This data shows when people visit different types of places. However, places are identified only by an ID and have no name.

B. Google Local Data for New York

This file [4] contains 272,189 businesses across New York State. For each business it gives the name, a list of categories, the average star rating, the number of reviews, the price level ($ to $$$$), weekly opening hours and whether the business is closed. It provides the names and quality information that the Foursquare data does not have.

C. Brooklyn Bridge Pedestrian Counts

The City of New York placed an automatic counter on the bridge walkway [5]. The dataset gives the number of pedestrians for each hour from October 2017 to December 2019, split into people walking towards Manhattan and towards Brooklyn. It also includes the weather, temperature and rainfall for each hour, and a note when a special event took place. It is the only dataset in this project with real crowd counts at a well-known landmark.

Because the three datasets come from different years, they are not used to describe how busy a place is today. They are used to learn general patterns, for example that parks are busiest in the morning. Because the three datasets come from different years, they are not used to describe how busy a place is today. They are used to learn general patterns, for example that parks are busiest in the morning. Each source also answers a different part of the planning problem: check-ins give visit timing and sequences, business records add names and practical details, and the bridge counter provides hourly crowd measurements with weather. The sources are combined for these complementary roles, not treated as if they were collected at the same time.

IV. SYSTEM OVERVIEW

The work in Phase 1 follows a simple pipeline with four stages. First, the three raw files are loaded and cleaned separately. Second, new features such as local hour, weekend flag, season and minutes to sunset are added, and the Foursquare places are matched to Google places to get names, ratings and opening hours. Third, the cleaned data is studied through thirteen analyses. Fourth, the results are turned into two tables, one row per place and one row per user, which the planner uses to score places against the form answers.

All work is done in Python in a single Jupyter notebook. The main libraries are pandas and NumPy for data handling, Matplotlib and seaborn for charts, SciPy for statistical tests and scikit-learn for the nearest-place search and for clustering. A separate download script fetches the three datasets from their public websites, so the full pipeline can be run again from the start. All work is done in Python in a single Jupyter notebook. The main libraries are pandas and NumPy for data handling, Matplotlib and seaborn for charts, SciPy for statistical tests and scikit-learn for the nearest-place search and for clustering. A separate download script fetches the three datasets from their public websites, so the full pipeline can be run again from the start. Keeping the cleaning and analysis steps in the notebook also makes it easier to trace a final result back to the data and the choices made along the way.

V. DATA CLEANING

A small logging function records the number of rows before and after every step, so the effect of each step can be seen clearly. Table II is produced directly from this log. The steps are described below in the same order as in the code. Overall, 89,112 of the 227,428 Foursquare check-ins (39.2%) and 41,164 of the 272,189 Google places (15.1%) remain after cleaning. Most of the Google rows are removed simply because they are outside New York City.

A. Loading the Data and Fixing Column Names

The Google file is too large to load into memory at once, so it is read in parts of 50,000 rows. From each part, only the needed columns and only the places inside a box around the five boroughs of New York City are kept. This removes 156,242 places located outside the city. After loading, all column names are changed to one naming style (for example,

userId becomes user_id), IDs are stored as text and counts are stored as numbers (Step 1).

B. Duplicates, Time Zones and Text Problems

Step 2 removes 250 check-ins that are exact copies, 608 Google places that appear more than once (the copy with the most reviews is kept) and one repeated hour in the bridge data. Duplicates must be removed because they make a place or an hour appear more popular than it really is.

Step 3 deals with time. Foursquare stores times in UTC, so a check-in made at 7 PM in New York is stored as 11 PM. Without correction, every "best time to visit" result would be wrong by four or five hours. All times are therefore converted to New York local time using the standard time zone rules, which also handle daylight saving time. The file also contains its own time offset column. This column disagrees with the converted time in 468 rows (0.21%), and some of these rows even contain offsets from Asian time zones. For this reason, the official time zone rules are used instead of the stored offset.

Step 4 fixes text problems. Some category names are broken because of a wrong character encoding; for example, the word Café appears as unreadable characters, which splits one category into several. A total of 332 such labels are repaired. In the bridge data, weather labels such as clear-day and clear-night are stored separately, even though the time already shows whether it is day or night. These labels are merged into a single label such as clear. In the Google data, some prices are written with the Korean won sign instead of the dollar sign. These are changed to dollar signs, because they mean the same price level, and without this fix these places would lose their price.

C. Removing Invalid or Irrelevant Rows

Step 5 removes 9,049 check-ins with locations outside New York City and 12,922 Google places that are permanently or temporarily closed. Recommending a closed place would be one of the most serious errors the app could make, so this step is important.

Step 6 removes the largest amount of data. Many check-ins are at places that a tourist would not visit, such as homes, offices, subway stations, gyms and schools. A set of keyword rules removes these places and groups the remaining categories into seven themes: Landmarks & Views, Museums & Arts, Parks & Outdoors, Entertainment, Food & Drink, Nightlife and Shopping. Some names can be misleading. For example, "Beer Garden" contains the word "garden" but is actually a bar, so special cases are added for such names. The code also prints the most common categories that do not match any rule, so that missing rules can be found and added. After this step, 105,811 check-ins and 41,164 Google places remain.

D. Opening Hours, Prices and Missing Values

Some Google fields are stored in a form that is difficult to use directly (Step 7). Categories are stored as lists, the price is written as dollar signs, and opening hours are written as text such as "11AM–3PM, 5–10PM". The opening hours are the most difficult field to process. They contain different dash characters, special space characters, two time ranges in one day, closing times after midnight, and phrases such as "Open 24 hours" and "Closed". A small parser is written for this field and tested on these difficult cases. It converts the text into numbers such as the usual opening hour, the usual closing hour and whether the place is open after 8 PM. This information is needed so that the app does not suggest visiting a place at 8 PM when it closes at 6 PM.

For missing values (Step 8), rows are not simply deleted whenever a value is blank. Instead, a separate rule is chosen for each column. In the Google data, about half of the places have no description and no price level, and about one in eight has no opening hours. For the bridge data, a complete list of every hour between the first and last record is created, which gives 19,728 hours and shows that 3,672 of them have no count. Short gaps of up to three hours in temperature and rainfall are filled, because these values change slowly. Pedestrian counts, however, are never filled in, since that would mean inventing data. Parks with no price information are given a price level of zero, because public parks are free.

E. Noise from Users and from the Sensor

Users sometimes check in to the same place several times within a few minutes. In Step 9, repeated check-ins by the same user at the same place within one hour are counted as one visit, which removes 778 rows.

Step 10 looks for check-ins that are physically impossible. The speed a user would need to travel between two consecutive check-ins is calculated. For most pairs the speed is very low (the median is 0.15 km/h, because people often stay in one area), but some users would need more than 1,500 km/h, which is faster than an airplane. Check-ins with a speed above 150 km/h, or with two places more than 2 km apart at the same second, are removed. This removes 566 rows.

Three checks are applied to the bridge counter in Step 11. First, the total count should equal the Manhattan count plus the Brooklyn count; two hours where this is not true are corrected. Second, one full day has a total of zero, which indicates that the sensor was switched off rather than that nobody used the bridge, so that day is set to missing. Third, 194 hours with very high counts are marked. These hours are not deleted, because many of them are real events such as New Year's Eve and summer fireworks.

F. Inactive Users, Time Features and Name Matching

A user with only two check-ins gives very little information about preferences, and a place with only two visits does not have a reliable "best hour". Step 12 keeps only users with at least five check-ins and places with at least three. Removing a place can push a user below five check-ins, so the filter is repeated until no further change happens. The process

TABLE II
ROWS REMOVED BY THE MAIN CLEANING STEPS

Step	Data	Before	After
2 Duplicates	Foursquare	227,428	227,178
5 Outside NYC	Foursquare	227,178	218,129
6 Non-travel categories	Foursquare	218,129	105,811
9 Repeated check-ins	Foursquare	105,811	105,033
10 Impossible movement	Foursquare	105,033	104,467
12 Inactive users/places	Foursquare	104,467	89,112
5 Outside NYC	Google	272,189	115,947
2 Duplicates	Google	115,947	115,339
5 Closed places	Google	115,339	102,417
6 Non-travel categories	Google	102,417	41,164

finishes after three rounds, leaving 1,074 users, 8,742 places and 89,112 check-ins.

Step 13 adds time features such as the hour, day of the week, weekend flag, month and season. Sunrise and sunset times are also calculated for every date using the formulas published by NOAA [6]. When checked against published sunset times for New York, the results are within about two minutes. With these values, each check-in receives the number of minutes before or after sunset and a label: daytime, golden hour (up to 90 minutes before sunset), dusk (up to 60 minutes after sunset) or night.

Step 14 adds names to the Foursquare places. For each Foursquare place, the closest Google place with the same theme is searched for. A larger distance limit is used for parks (400 m) because parks cover a large area, and a small limit is used for restaurants and bars (40 m) because they are close to each other. About 72.9% of places receive a name in this way. The match rate is highest for Food & Drink (77.8%) and lowest for Landmarks & Views (51.8%), because many landmarks such as statues and plazas are not listed as businesses on Google. Step 15 removes columns that are no longer needed and saves the cleaned files.

VI. EXPLORATORY DATA ANALYSIS

Thirteen analyses are carried out on the cleaned data. For each analysis, the main finding and its effect on the design of the app are recorded. The most useful results are described in this section, and some of the key numbers for each theme are collected in Table III.

A. Uneven Popularity

Most counts in the data are very uneven. A place has an average of 10.2 check-ins, but the median is only 6, because a small number of very popular places raise the average. Users are also uneven: the average user has 83 check-ins, the median user has 68, and the most active user has 998. Google reviews show the same pattern, with an average of 265 reviews and a median of 88. The top 10% of places receive 42.0% of

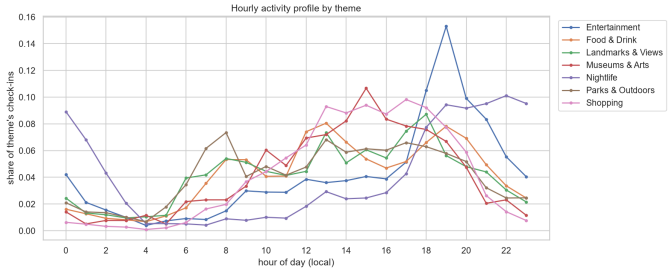


Fig. 1. Share of each theme's check-ins by hour of the day. Each line adds up to one, so themes of different sizes can be compared.

all check-ins. Food & Drink alone accounts for 51.8% of all check-ins, while Museums & Arts accounts for only 0.9%. A likely reason is that most Foursquare users are residents of New York rather than tourists.

This result affects how places will be ranked. If places are ranked only by popularity, every user will receive the same ten famous places. For this reason, medians are used to describe typical values, and the logarithm of popularity will be used in the model so that very popular places do not dominate the results. This result affects how places will be ranked. If places are ranked only by popularity, every user will receive the same ten famous places. For this reason, medians are used to describe typical values, and the logarithm of popularity will be used in the model so that very popular places do not dominate the results. This keeps popularity as one useful signal while leaving room for a less famous place to rank well when it better matches a traveler's interests and preferred time.

B. Busiest Hours for Each Theme

Fig. 1 shows the share of each theme's check-ins at each hour of the day. Shares are used instead of raw counts so that small themes such as museums can be compared with large themes such as food. Each theme has a clear pattern. Parks are busiest at 8 AM, food at 1 PM, museums at 3 PM, shopping at 5 PM, landmarks at 6 PM, entertainment at 7 PM and nightlife at 10 PM. Entertainment has the sharpest peak, with 15.3% of all its check-ins in the 7 PM hour, which matches the start time of most shows and games. This is the most direct answer to the question "when should I go?", and the planner uses it to choose the time of each stop.

The same data also shows the quietest daytime hours. For most themes these are in the morning between 8 AM and 11 AM, so a traveler who wants to avoid crowds can be sent to popular places early in the day.

Weekdays and weekends are also compared. For most themes, the typical hour changes by only one hour. Nightlife is an exception: the median check-in time moves from 7 PM on weekdays to 3 PM on weekends, which may be explained by weekend brunch and afternoon drinks. As a result, the planner will use separate timings for weekdays and weekends. Weekdays and weekends are also compared. For most themes, the typical hour changes by only one hour. Nightlife is an

TABLE III
KEY NUMBERS FOR EACH THEME (GAP: MEDIAN TIME TO THE NEXT STOP; NAMED: SHARE OF PLACES MATCHED TO A GOOGLE NAME; WEEKEND: SHARE OF CHECK-INS ON SATURDAY OR SUNDAY)

Theme	Peak hour	Gap to next	Named	Weekend
Landmarks & Views	6 PM	48 min	51.8%	28.7%
Museums & Arts	3 PM	119 min	58.3%	32.3%
Parks & Outdoors	8 AM	65 min	71.6%	33.8%
Entertainment	7 PM	160 min	64.0%	35.6%
Food & Drink	1 PM	118 min	77.8%	30.2%
Nightlife	10 PM	126 min	63.1%	41.3%
Shopping	5 PM	59 min	72.2%	32.7%

exception: the median check-in time moves from 7 PM on weekdays to 3 PM on weekends, which may be explained by weekend brunch and afternoon drinks. As a result, the planner will use separate timings for weekdays and weekends. This matters because a single average schedule could hide the later weekend pattern, especially for nightlife, and could place a stop at a time that is typical only on a different kind of day.

C. Day and Hour Pattern

A heatmap of check-ins by day of the week and hour of the day is also made in the notebook. Activity is highest in the afternoon and early evening, and the single busiest time of the whole week is Friday at 7 PM, with 1,351 check-ins. On weekdays there is a clear lunch peak and an evening peak. On Saturday and Sunday, activity starts later in the morning, stays high through the afternoon and continues later into the night. Saturday has the most check-ins of any day (16,037) and Thursday has the fewest (10,813).

Weekends are 2 of the 7 days, or about 29% of the week. If a theme has more than 29% of its check-ins on weekends, it is more of a weekend activity. As shown in Table III, 41.3% of Nightlife and 35.6% of Entertainment check-ins happen on weekends, while Landmarks & Views (28.7%) is visited evenly across the week. This supports the decision to plan weekdays and weekends differently.

D. The Sunset Effect

We expected the sunset analysis to give one of the strongest results, but that was not what we found. For each theme and light period, the "lift" is calculated. Lift is the share of a theme's check-ins in a period divided by the share of all check-ins in that period. A lift of 1 means normal, a value above 1 means the theme is more common than usual in that period, and a value below 1 means it is less common. The results are shown in Fig. 2 (bottom). Entertainment has the highest golden hour lift (1.56), and nightlife is very high at night (2.01). However, landmarks (0.93) and parks (0.86) are slightly below normal at golden hour, which is the opposite of what was expected.

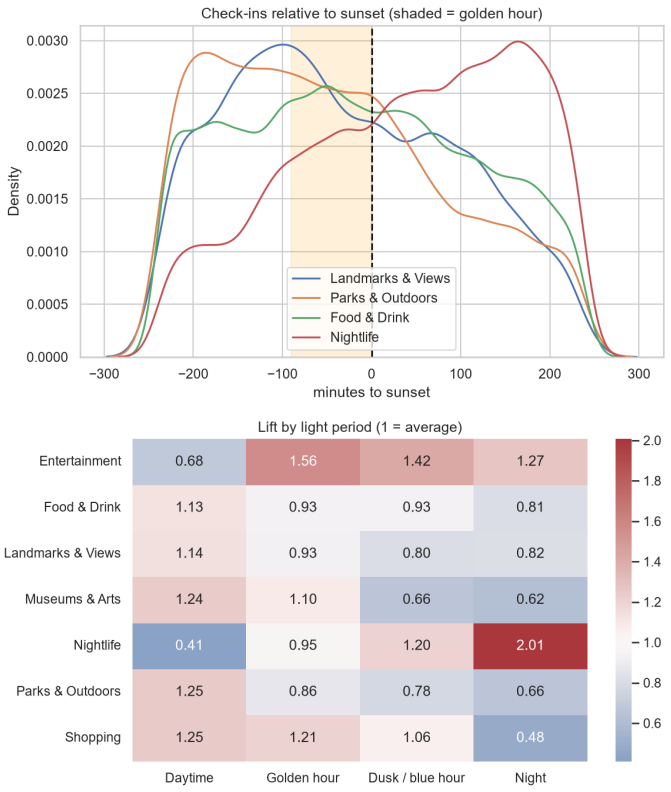


Fig. 2. Top: check-ins relative to sunset for four themes (the shaded band is golden hour). Bottom: lift of each theme in each light period, where 1 means normal.

A further problem was then identified. In this data, "minutes to sunset" and "hour of the day" are almost the same variable (correlation 0.98). Therefore, if a theme appears popular near sunset, it may simply be popular in the early evening. To separate the two, the change in sunset time across the year is used. Sunset moves by more than three hours during the year, so 7 PM is golden hour in June but already dark in December. Check-ins in the same clock hours (from 4 PM to 9 PM) are compared between days when that hour is near sunset and days when it is not, and the hours are combined into a single Mantel–Haenszel odds ratio. A value of 1 means that sunset makes no difference. The results are 1.04 for landmarks and parks, 0.97 for food and 0.92 for nightlife. Once the clock hour is fixed, sunset has almost no effect on where people check in.
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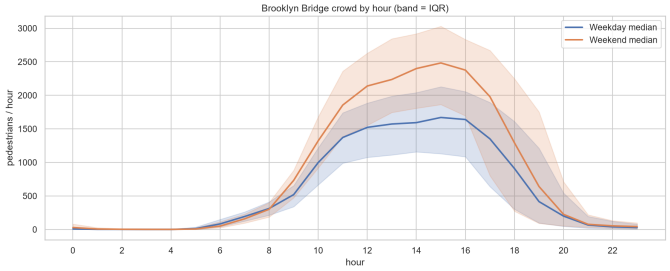


Fig. 3. Brooklyn Bridge pedestrians per hour on weekdays and weekends. The lines are medians and the bands show the middle half of the values.

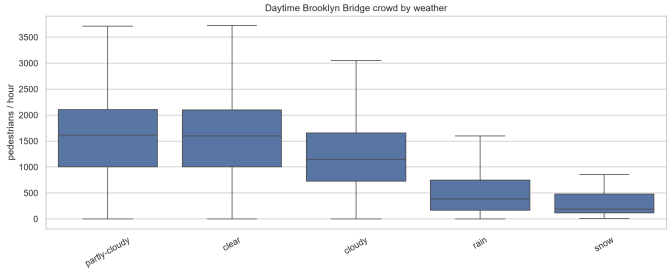


Fig. 4. Daytime Brooklyn Bridge counts for each weather type (extreme values hidden).

parks, 0.97 for food and 0.92 for nightlife. Once the clock hour is fixed, sunset has almost no effect on where people check in. In other words, the early-evening timing explains most of the apparent pattern; the analysis does not find a separate increase that can confidently be attributed to sunset itself.

The list of places with the highest share of sunset visits also shows the problem. The top places include small playgrounds, places with no name and even a parking lot, each with only 20 to 45 check-ins. These are not real sunset viewpoints; they appear at the top only because their numbers are small.

A likely explanation is that Foursquare users in 2012 were mostly residents following a daily routine, not tourists trying to see a view. The sunset feature can still be useful. We can calculate sunset time for any travel date, and if a traveler asks for sunset photos, the planner can suggest a viewpoint during golden hour. However, the data does not prove that people prefer viewpoints at sunset. The design is therefore changed so that sunset is used as a scheduling rule for users who request it, and not as a value learned from popularity.

E. Location of Activity

Density maps for each theme are included in the notebook. Most activity is found in Lower and Midtown Manhattan. Using a simple box around Manhattan, about 67% of all check-ins fall inside it, while Brooklyn and Queens have smaller clusters. Parks form separate clusters, and nightlife is concentrated in a few neighborhoods. Among the most visited named places, chains such as Starbucks (2,514 check-ins) and Dunkin' (868 check-ins) appear at the top, together with real attractions such as Madison Square Park, One Times Square and the Bryant Park area. Since tourists rarely travel to New York to visit such chains, their scores will be reduced in the planner. The maps also suggest that a day plan should stay inside one or two nearby areas, so that the traveler does not spend most of the day on the subway.

F. Brooklyn Bridge Crowds by Hour

Fig. 3 shows the median number of pedestrians on the bridge for each hour, with a band covering the middle half of the values. Both weekdays and weekends reach their peak at 3 PM, but weekends are much busier, with a median of 2,483 pedestrians per hour compared with 1,670 on weekdays. At 8 PM the median is about 200 on weekdays and 226 on weekends, which is around 88% to 91% lower than the 3 PM peak. Early morning is also quiet, with a median of about 300 people in the 8 AM hour. The averages tell the same story: at 3 PM there are on average 1,637 people per hour on weekdays and 2,354 on weekends, while at 8 PM there are only 379 and 511, which is about 77% fewer. The crowd also changes with the season, from an average of 892 people per hour in summer to 438 in winter. This gives the planner a measured reason to suggest an early morning or evening walk across the bridge instead of an afternoon walk.

The busiest single hours in the data are clearly linked to events. For example, the 10 AM hour on November 10, 2019 had 3,312 people, and several summer evenings in 2018 had more than 1,000 people between 8 PM and 10 PM. This supports the decision in Step 11 to keep these hours instead of deleting them.

G. Effect of Weather on Crowds

This analysis checks whether bad weather keeps people away. Only daytime hours (10 AM to 8 PM) are used, so that night hours with very few people do not affect the result. Weather types with at least 30 hours of data are kept (Fig. 4). Since the counts are not normally distributed, the Kruskal–Wallis test [7] is used, which compares groups using ranks instead of raw values. The result is H = 772.2 with a p-value of about 8×10^{-166}, so the difference is clearly significant. The effect size, measured as epsilon squared, is 0.12, which is a medium effect. On a clear afternoon the median is 1,598 pedestrians per hour, while on a rainy afternoon it is only 386, about 76% fewer. Snow gives the lowest median of all, at 191.

Rain and snow are more common in colder months, so the result could simply be a seasonal effect. To check this, clear and rainy hours are compared within the same season. The effect remains: in summer the median falls from 1,804 (clear) to 860 (rain), and in winter it falls from 992 to 252. Temperature also has an effect, with a Spearman correlation of 0.51 between daytime temperature and crowd size. The weather forecast should therefore be an input to the planner, and on rainy days outdoor stops should be replaced with indoor ones such as museums and food places.

H. Fair Ratings

Almost all Google ratings lie between 4 and 5 stars (average 4.24, median 4.3), so the raw rating does not separate good places from average ones very well. It also favors places with very few reviews, since a place with three 5-star reviews appears better than a place with 5,000 reviews and 4.7 stars. To correct this, a Bayesian average is used:

$$s = \frac{v}{v+m}R + \frac{m}{v+m}C, \tag{1}$$

where v is the number of reviews of the place, R is its rating, C is the average rating of all places and m is the median number of reviews (88). In simple terms, a place with few reviews is pulled towards the overall average, while a place with many reviews keeps a score close to its own rating. Before this correction, the top ten landmarks were all rated 5.0, and most of them were small tour companies with about 100 to 250 reviews. After applying (1), well-known places such as the 9/11 Memorial South Pool (4,400 reviews), St. Patrick's Cathedral (9,998 reviews) and Bow Bridge in Central Park (1,737 reviews) move into the top ten.

Ratings also differ a little between themes. About 80% of all places are rated 4.0 or more. Museums & Arts have the highest median rating (4.6), followed by Landmarks & Views and Entertainment (4.5), Parks & Outdoors and Nightlife (4.4), and Food & Drink and Shopping (4.3). The number of reviews is also very different between themes: the median Food & Drink place has 118 reviews, while the median museum has only 17. This is another reason to use the rating together with the number of reviews.

This comparison also revealed an error in the theme rules. Some escape rooms and an axe throwing venue were placed in the Landmarks theme, because one of their secondary Google categories is "Tourist attraction". This rule will be corrected in Phase 2.

I. Next Stop After Each Theme

To decide the order of stops in a day, the next activity after each check-in is studied. Pairs of check-ins by the same user on the same day (33,065 pairs) are used to calculate the probability of each next theme given the current theme (Fig. 5). Food & Drink is the most likely next stop after every theme, ranging from 0.38 after landmarks to 0.55 after another food stop. Nightlife is often followed by more nightlife (0.39), and 15% of landmark visits are followed by a park. Museums are almost never the next stop, with a probability of about 0.01 from most themes.

The typical time gap before the next stop is also measured, and it is listed in Table III. The gap is short after landmarks (48 minutes) and shopping (59 minutes), and long after entertainment (160 minutes), which makes sense because a show or a game takes a few hours. The transition table will be used to order the stops, and the time gaps will be used to decide how much time to give each stop. The typical time gap before

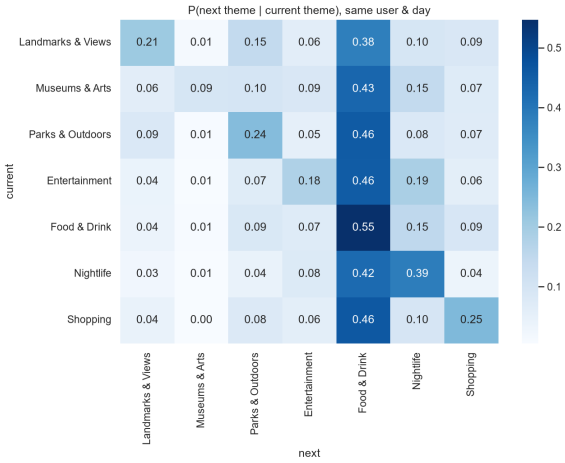


Fig. 5. Probability of the next theme given the current theme, for check-ins by the same user on the same day.

the next stop is also measured, and it is listed in Table III. The gap is short after landmarks (48 minutes) and shopping (59 minutes), and long after entertainment (160 minutes), which makes sense because a show or a game takes a few hours. The transition table will be used to order the stops, and the time gaps will be used to decide how much time to give each stop. These are patterns from residents' recorded check-ins, so they are a starting point for arranging a day rather than a rule that every tourist must follow. The planner should combine them with the traveler's choices and practical limits such as opening hours and distance.

J. Possibility of Personalization

A table of users against places is created, with a mark wherever a user visited a place. The table has $1{,}074 \times 8{,}742$ cells, and 99.53% of them are empty. With such sparse data, methods that learn from similar users do not work well at the level of individual places. In addition, a new traveler has no history at all. Personalization is therefore done at the theme level: each user is described by seven numbers, one per theme, and the form fills in these numbers directly. This is the main reason the app begins with a form.

Correlations between the bridge features are also checked. The Manhattan and Brooklyn direction counts are almost identical to the total (correlation above 0.95), so they are not used to predict the total. The hour and the minutes to sunset are also almost the same (0.98), which led to the additional sunset test described in Section VI-D.

A simple correlation with the number of pedestrians was also checked for the other bridge features. Temperature has the strongest link with the crowd (0.37), followed by hour of the day (0.30) and weekend (0.12). Month has a very small link (0.07), rainfall has a small negative link (-0.10), and the event flag shows almost no link (0.00). Based on this, a future crowd model will use temperature, hour, weekend, month and rainfall as inputs.

TABLE IV
FORM QUESTIONS AND THE DATA FEATURES THEY USE

Question	Feature in the data
Interests (pick any)	theme of the place
Preferred time of day	share of visits in each part of the day
Want sunset spots?	sunset lift, sunset time of the date
Feeling about crowds	popularity of the place
Pace of the day	stops per day, time between stops
Budget per stop	Google price level
Travel date	sunset time, season, weather

VII. LINKING THE FORM TO THE DATA

The form contains seven questions. Table IV lists each question and the data feature it is linked to.

A. Place Profiles

In Step 16, one row is created for each of the 8,742 places with all the values the form needs: theme, share of visits in the morning, afternoon, evening and night, share of weekend visits, sunset lift, popularity, best hour, and, for named places, the Bayesian rating, price and opening hours. For places with fewer than 15 check-ins the best hour is not reliable, so the busiest hour of the whole theme is used instead. In Step 16, one row is created for each of the 8,742 places with all the values the form needs: theme, share of visits in the morning, afternoon, evening and night, share of weekend visits, sunset lift, popularity, best hour, and, for named places, the Bayesian rating, price and opening hours. For places with fewer than 15 check-ins the best hour is not reliable, so the busiest hour of the whole theme is used instead. This avoids treating a few visits as a stable pattern for an individual place while still giving the planner a reasonable time to try. The profile keeps the place-level information where there is enough data and falls back to the broader theme pattern where there is not.

B. User Profiles

In Step 17, a real survey of a thousand travelers is not possible, so the answers each Foursquare user would give are estimated from their actual behavior. For example, a user's interests are the themes that make up at least 15% of their check-ins, and their crowd answer is based on how popular the places they visit are. The estimated answers look reasonable: 36% of users would choose evening as their preferred time, 33% afternoon, 19% morning and 13% night. These estimated answers will be used to test the planner in the next phase: answers are estimated from a user's earlier check-ins, a plan is created, and the plan is compared with the places the user visited later. One limitation is that most of these users are residents, so the profiles describe general preferences rather than tourist behavior. In Step 17, a real survey of a thousand travelers is not possible, so the answers each Foursquare user would give are estimated from their actual behavior. For example, a user's interests are the themes that make up at least 15% of their check-ins, and their crowd answer is based on how popular the places they visit are. The estimated answers look reasonable: 36% of users would choose evening as their preferred time, 33% afternoon, 19% morning and 13% night. These estimated answers will be used to test the planner in the next phase: answers are estimated from a user's earlier check-ins, a plan is created, and the plan is compared with the places the user visited later. One limitation is that most of these users are residents, so the profiles describe general preferences rather than tourist behavior. These profiles are useful for checking that the form-to-planner pipeline works, but they should not be presented as survey responses or as proof that the same choices represent visitors to New York.

C. Usefulness of Each Question

A good question should receive different answers from different people and should not repeat another question. The largest overlap is between preferring evenings and making sunset visits (correlation 0.78), which means the sunset question adds little information for users who already chose "evening". Interest in nightlife and activity at night are also related (0.55). The budget question also shows little variation, since among users with known prices, 62% would answer $$ and only 1% would answer $$$ or more. All questions are kept for now, but these questions may be merged or made optional later.

D. Personas

K-means clustering is also tried to group users, with the aim of offering ready-made types such as "Foodie" or "Night owl" that fill in the form with one tap. The quality of the groups is measured with the silhouette score [8], which ranges from -1 to 1, where higher values mean clearer groups. The best score is only 0.13, with seven groups. The largest groups are users who are active in the evening (331 users), nightlife users (249) and morning users (236), while the smallest groups are landmark lovers (46) and very fast-paced users (36). Since the score is low, the groups are weak and overlap heavily. User preferences therefore do not fall into clear categories, and the separate questions are kept instead of personas. K-means clustering is also tried to group users, with the aim of offering ready-made types such as "Foodie" or "Night owl" that fill in the form with one tap. The quality of the groups is measured with the silhouette score [8], which ranges from -1 to 1, where higher values mean clearer groups. The best score is only 0.13, with seven groups. The largest groups are users who are active in the evening (331 users), nightlife users (249) and morning users (236), while the smallest groups are landmark lovers (46) and very fast-paced users (36). Since the score is low, the groups are weak and overlap heavily. A label such as "Foodie" might sound clear, but the measured behavior does not separate users into reliable, distinct types. Asking people directly about their interests is a better fit for this version of the planner. User preferences therefore do not fall into clear categories, and the separate questions are kept instead of personas.

TABLE V
MAIN PLAN RETURNED FOR THE SAMPLE USER

Time	Stop	Theme
8:00 AM	Enid A. Haupt Conservatory	Parks
1:00 PM	Caffè Panna	Food
2:15 PM	Uncle Gussy's	Food
5:25 PM	Elizabeth Street Garden (sunset)	Parks
6:40 PM	Ferry (unnamed place)	Landmarks
7:55 PM	William Shakespeare Statue	Landmarks

VIII. TESTING A SIMPLE PLANNER

A simple planner is written to check that all the tables work together. It gives each place a score by adding how well the place matches the user's interests, time of day, sunset choice, rating and crowd choice, and it subtracts points if the place is over budget. The highest scoring places are placed at their best hour, one viewpoint is moved to the golden hour of the travel date, and stops are kept at least 75 minutes apart. When the user dislikes a stop, the weight of its theme is reduced and a backup plan is created from the remaining places. A simple planner is written to check that all the tables work together. It gives each place a score by adding how well the place matches the user's interests, time of day, sunset choice, rating and crowd choice, and it subtracts points if the place is over budget. The highest scoring places are placed at their best hour, one viewpoint is moved to the golden hour of the travel date, and stops are kept at least 75 minutes apart. When the user dislikes a stop, the weight of its theme is reduced and a backup plan is created from the remaining places. At this stage, the score is mainly a way to check that the user's answers can be combined with the place profiles; it is not yet a learned or validated recommendation model.

The planner is tested with a sample user who likes landmarks, food and parks, prefers evenings, wants sunset spots, wants a balanced day, has a $$ budget and travels on October 10, 2026, when sunset is at 6:25 PM. The main plan returned by the planner is shown in Table V. It places a garden at 5:25 PM as the sunset spot, which shows that the complete flow works from the form answers to a timed plan. The planner is tested with a sample user who likes landmarks, food and parks, prefers evenings, wants sunset spots, wants a balanced day, has a $$ budget and travels on October 10, 2026, when sunset is at 6:25 PM. The main plan returned by the planner is shown in Table V. It places a garden at 5:25 PM as the sunset spot, which shows that the complete flow works from the form answers to a timed plan. The example is useful as an integration check: interests, budget, preferred time and travel date all reach the output. It is not yet evidence that the route is convenient, since the issues below show that some practical constraints still need to be added.

A closer look at the plan, however, shows several problems that will be addressed in Phase 2:

- Opening hours are not checked yet. The garden is scheduled at 5:25 PM although it closes at 5 PM, and the conservatory, which has no opening hours in the data, is scheduled at 8 AM.
- Distance is not considered. The plan moves from the Bronx to lower Manhattan and back to Central Park, a total of about 29 km in one day.
- The next-stop table is not used yet, so two food stops appear one after another at lunch time.
- One stop has no name, because its Foursquare place could not be matched to a Google place. Unnamed places should not be shown to a traveler.
- Some selected places have only one to four different visitors, so their sunset values are based on very few check-ins and are not reliable.
- The penalty for a disliked theme is too small, so the backup plan still contains two parks after the user dislikes parks.

IX. LIMITATIONS

The Foursquare data is from 2012 and 2013 and covers only about ten and a half months. It has no data for March, and it includes the weeks around Hurricane Sandy, so seasonal results from it should be treated with care. The Google data is from around 2021 and the bridge data from 2017 to 2019, so the three sources do not describe the same period. The name matching uses location only, so it is not fully accurate. For example, 883 Google places are matched to more than one Foursquare place, and one small place receives 772 check-ins from only two users, which suggests the area around a user's home. The bridge counter covers only one landmark, so crowd levels at other landmarks have to be estimated from check-ins. Finally, the theme rules are written by hand, and, as the ratings analysis showed, a few categories are placed in the wrong theme. The Foursquare data is from 2012 and 2013 and covers only about ten and a half months. It has no data for March, and it includes the weeks around Hurricane Sandy, so seasonal results from it should be treated with care. The Google data is from around 2021 and the bridge data from 2017 to 2019, so the three sources do not describe the same period. The name matching uses location only, so it is not fully accurate. For example, 883 Google places are matched to more than one Foursquare place, and one small place receives 772 check-ins from only two users, which suggests the area around a user's home. The bridge counter covers only one landmark, so crowd levels at other landmarks have to be estimated from check-ins. Finally, the theme rules are written by hand, and, as the ratings analysis showed, a few categories are placed in the wrong theme. The findings should therefore be read as results for the cleaned data and these current category rules. Updating the source data and reviewing the rules could change some counts and recommendations, so both will need to be checked again before the planner is used for real trips.

X. Conclusion and Future Work

In Phase 1, three very different public datasets were cleaned and joined, and the results show that they contain useful information about timing. Each type of place has its own busy hours, about two thirds of all check-ins are in Manhattan, nightlife and entertainment are more popular on weekends, rain reduces the Brooklyn Bridge crowd by about three quarters, and evenings on the bridge are around 90% quieter than the afternoon peak. The sunset effect is smaller than expected, and the design has been changed to reflect this result. Every question in the form is now linked to a feature in the data, and a first test planner already produces a timed plan and a backup plan.

In Phase 2, we will work on making the planner more practical. The main tasks are:

- Correcting the theme rules and removing chains and unnamed places from the list of stops;
- Adding checks for opening hours and keeping each day within a small area;
- Using the next-stop table and the typical gaps to order the stops and set their durations;
- Giving less weight to places with very few visits;
- Using the weather forecast to swap outdoor stops for indoor ones on rainy days;
- Building a simple crowd model from temperature, hour, weekend, month and rainfall;
- Testing the planner by comparing its plans with the places users actually visited later.

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